

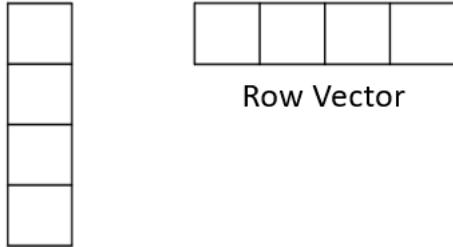
Reading: Matrix Mathematics

Estimated effort: 5 mins

You have seen that you can use Numpy package functions to perform different types of operations on arrays and matrices. In this reading, you will learn how these operations work mathematically.

1D Arrays : Vectors

A 1D array is often termed as a vector. Depending upon the orientation of the data, the vector can be classified as a row vector or a column vector. This is illustrated in the image below.



Row Vector

Column Vector

Mathematically, we can add, subtract, and take the product of two vectors, provided they are the same shape. The images below highlight the mathematical operations conducted on a pair of vectors.

$$\begin{array}{|c|} \hline A1 \\ \hline A2 \\ \hline A3 \\ \hline A4 \\ \hline \end{array} + \begin{array}{|c|} \hline B1 \\ \hline B2 \\ \hline B3 \\ \hline B4 \\ \hline \end{array} = \begin{array}{|c|} \hline A1+B1 \\ \hline A2+B2 \\ \hline A3+B3 \\ \hline A4+B4 \\ \hline \end{array}$$

Vector addition

$$\begin{array}{|c|} \hline A1 \\ \hline A2 \\ \hline A3 \\ \hline A4 \\ \hline \end{array} - \begin{array}{|c|} \hline B1 \\ \hline B2 \\ \hline B3 \\ \hline B4 \\ \hline \end{array} = \begin{array}{|c|} \hline A1-B1 \\ \hline A2-B2 \\ \hline A3-B3 \\ \hline A4-B4 \\ \hline \end{array}$$

Vector Subtraction

$$\begin{array}{|c|} \hline A1 \\ \hline A2 \\ \hline A3 \\ \hline A4 \\ \hline \end{array} \cdot \begin{array}{|c|} \hline B1 \\ \hline B2 \\ \hline B3 \\ \hline B4 \\ \hline \end{array} = \begin{array}{|c|} \hline A1 \cdot B1 \\ \hline A2 \cdot B2 \\ \hline A3 \cdot B3 \\ \hline A4 \cdot B4 \\ \hline \end{array}$$

Vector Product

All three of these operations are conducted on corresponding elements of individual vectors. The resulting array always has the same size as that of the two original vectors.

To a single vector, we can also add a constant (scalar addition), subtract a constant (scalar subtraction) and multiply a constant (scalar multiplication) to any vector. The images below illustrate these operations.

A1
A2
A3
A4

 $+ C =$

A1+C
A2+C
A3+C
A4+C

Scalar addition

A1
A2
A3
A4

 $- C =$

A1-C
A2-C
A3-C
A4-C

Scalar Subtraction

A1
A2
A3
A4

 $\times C =$

A1×C
A2×C
A3×C
A4×C

Scalar Product

2D Arrays : Matrices

A 2D array is also called a Matrix. These are typically rectangular arrays with data stored in different rows. All of the operations mentioned above are also applicable to the 2D arrays.

As illustrated in the images below, the dot product is carried out by multiplying and adding corresponding elements of rows of the first matrix with the elements of columns of the second matrix. As a result, the output matrix from the multiplication will have a modified shape.

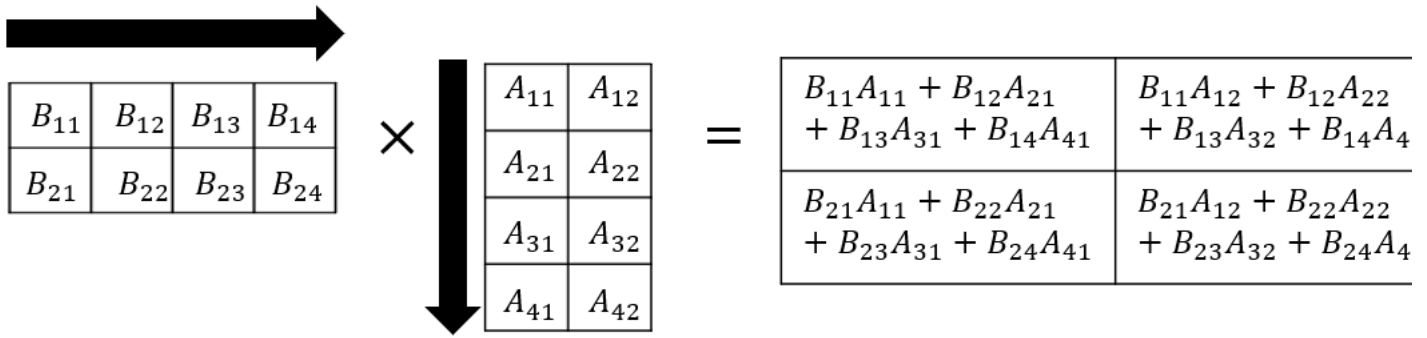
The general rule is that the dot product of an $m \times n$ matrix can be done only with an $n \times p$ matrix, and the resultant matrix will have the shape $m \times p$. In the example shown below, the 4×2 matrix is multiplied with the 2×4 matrix to generate a 4×4 matrix.

$\times$

$=$

$A_{11}B_{11}$ +	$A_{11}B_{12}$ +	$A_{11}B_{13}$ +	$A_{11}B_{14}$ +
$A_{12}B_{21}$	$A_{12}B_{22}$	$A_{12}B_{23}$	$A_{12}B_{24}$
$A_{21}B_{11}$ +	$A_{21}B_{12}$ +	$A_{21}B_{13}$ +	$A_{21}B_{14}$ +
$A_{22}B_{21}$	$A_{22}B_{22}$	$A_{22}B_{23}$	$A_{22}B_{24}$
$A_{31}B_{11}$ +	$A_{31}B_{12}$ +	$A_{31}B_{13}$ +	$A_{31}B_{14}$ +
$A_{32}B_{21}$	$A_{32}B_{22}$	$A_{32}B_{23}$	$A_{32}B_{24}$
$A_{41}B_{11}$ +	$A_{41}B_{12}$ +	$A_{41}B_{13}$ +	$A_{41}B_{14}$ +
$A_{42}B_{21}$	$A_{42}B_{22}$	$A_{42}B_{23}$	$A_{42}B_{24}$

In the reverse example, when 2×4 matrix is multiplied with the 4×2 one, the resultant will be a 2×2 matrix.



The diagram illustrates the multiplication of two matrices. On the left, a 2x4 matrix B is shown with elements $B_{11}, B_{12}, B_{13}, B_{14}$ in the first row and $B_{21}, B_{22}, B_{23}, B_{24}$ in the second row. A thick black arrow points to the right above this matrix. To its right is a multiplication symbol $\times$, followed by a 4x2 matrix A with elements A_{11}, A_{12} in the first column and A_{21}, A_{22} in the second column, and so on down to A_{41}, A_{42} . A thick black arrow points downwards to the right of this matrix. To the right of the multiplication is an equals sign $=$, followed by the resulting 2x2 matrix. The elements of the result matrix are calculated as follows:

$B_{11}A_{11} + B_{12}A_{21} + B_{13}A_{31} + B_{14}A_{41}$	$B_{11}A_{12} + B_{12}A_{22} + B_{13}A_{32} + B_{14}A_{42}$
$B_{21}A_{11} + B_{22}A_{21} + B_{23}A_{31} + B_{24}A_{41}$	$B_{21}A_{12} + B_{22}A_{22} + B_{23}A_{32} + B_{24}A_{42}$

Note: Dot product of a row vector with a column vector, with the same number of elements, would return a single scalar value. Dot product of a column vector with a row vector, will return a 2D matrix.

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